

ENGINEERING PROJECT PORTFOLIO

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A selection of design, simulation and build projects spanning automotive and high-performance engineering.

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Adaptive Electromagnetic Valve Timing (AEVT)

01

MEng Individual Project · Oct 2025 – May 2026

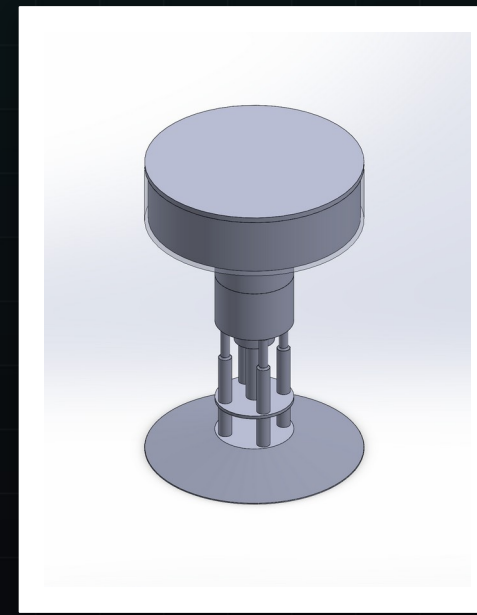
OVERVIEW

A camless engine valve-train concept taken from first principles to a working, electronically controlled prototype.

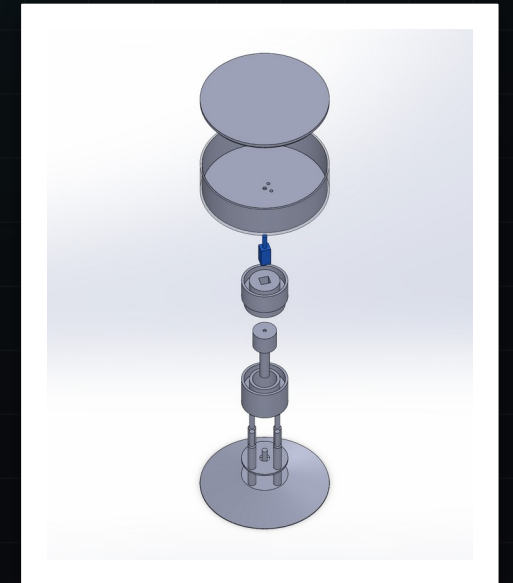
- Designed and built a solenoid-based cam-less valve actuation prototype as a fully variable alternative to conventional cam-driven systems, achieving up to 15 mm of valve lift with spring return.
- Implemented an Arduino control system with MOSFET switching and optocoupler isolation to generate variable-frequency PWM signals simulating engine throttle and speed.
- Completed end-to-end development across mechanical design, material selection, circuit design and testing, demonstrating electronically variable valve timing for improved efficiency and reduced pumping losses.



Build prototype



CAD assembly



Exploded view

Machine Learning-Based Adaptive Train Suspension

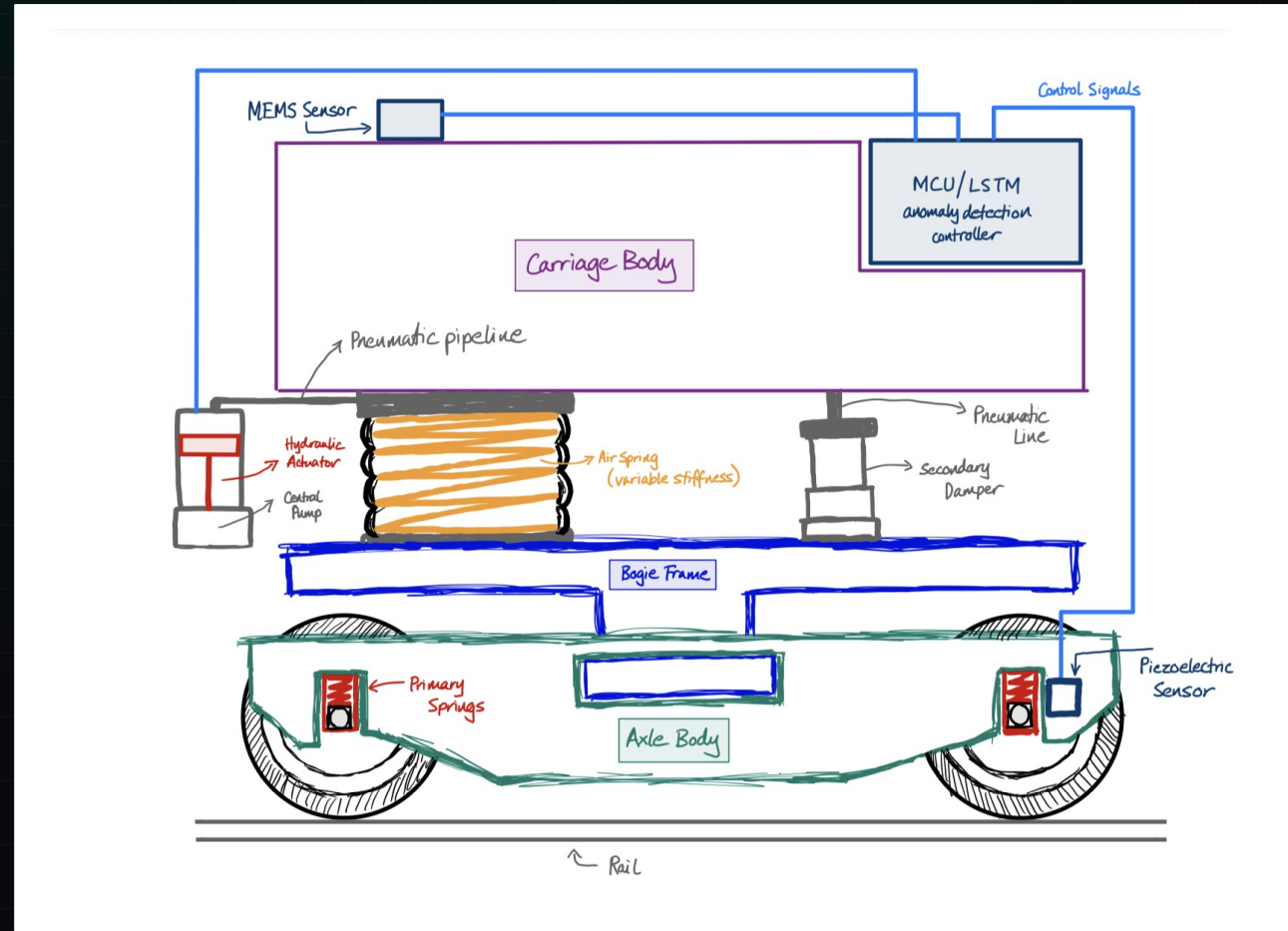
02

Group Project · Jan 2026 – Apr 2026

OVERVIEW

Bringing predictive, machine-learning fault detection to high-speed rail suspension.

- Proposed and simulated an ML-enhanced adaptive electronic suspension system for high-speed rail carriages, formulating a two-degree-of-freedom model and training an LSTM autoencoder on healthy vibration data to detect track anomalies in real time.
- Designed a coordinated safety response integrating rapid secondary damping adjustment and automatic speed reduction, shifting suspension from reactive control to predictive, data-driven fault detection.



System architecture and anomaly-detection sensing

Biofuel Storage - AC Adsorption Measurement System

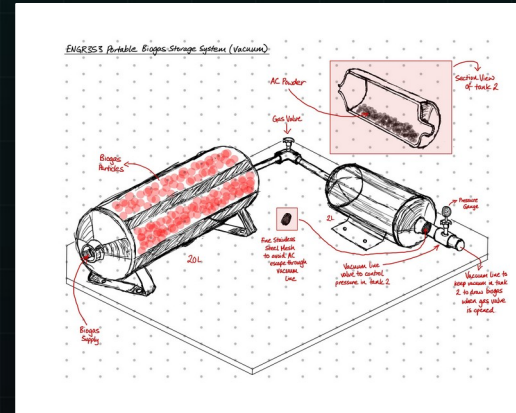
03

Group Design Project · Oct 2025 – Dec 2025

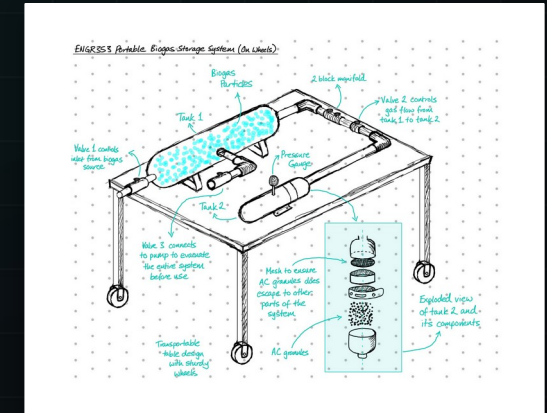
OVERVIEW

A field-portable rig for characterising activated-carbon methane storage on a tight budget.

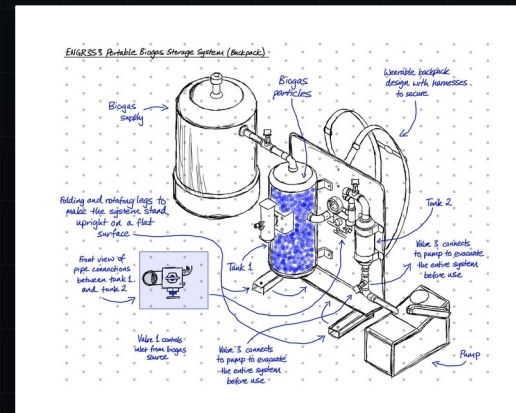
- Contributed to designing a portable, low-cost system for measuring activated carbon adsorption fill-up behaviour in biogas transport, enabling methane storage testing at pressures up to 12 bar within a £2,000 budget.
- Produced design sketches and led concept selection, supporting a dual-container pressure vessel arrangement sized to test 200 g of AC and transportable over rough terrain by a single person.



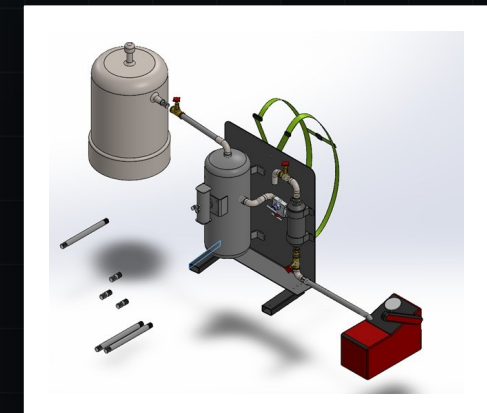
Vacuum-line design concept



Transportable wheeled design



Design iteration



Final CAD arrangement

Autonomous Track-Following Robot

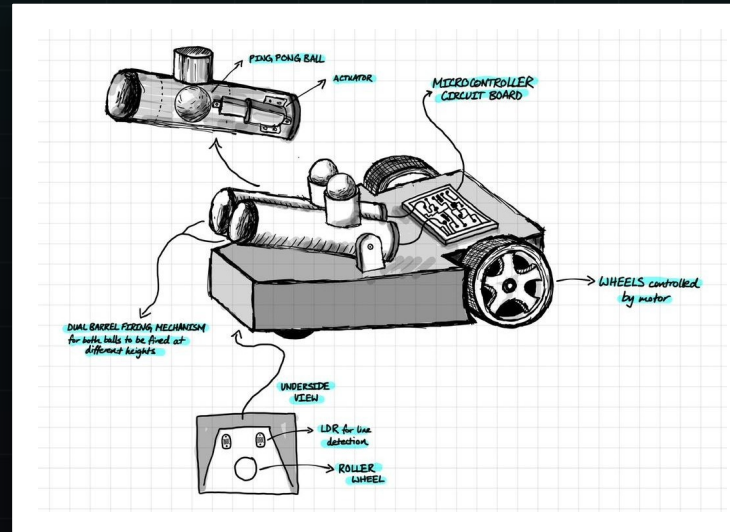
04

Group Project · Oct 2024 – Mar 2025

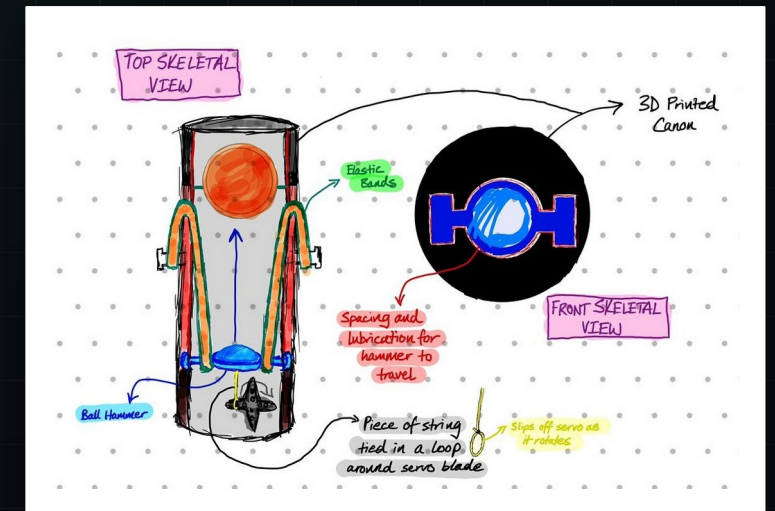
OVERVIEW

An autonomous line-following robot with an onboard ball-targeting and firing system.

- Designed, built and programmed an autonomous robot to follow a 15 m track and accurately launch two ping-pong balls into target zones at different distances, using LDR Wheatstone bridge sensing and PID-tuned motor control.
- Engineered a servo-actuated elastic band firing mechanism triggered on wall contact, and contributed to CAD modelling, PCB soldering and structured testing throughout development.



Chassis and sensor layout



Ball-launcher mechanism

IMechE Foundation Design Challenge

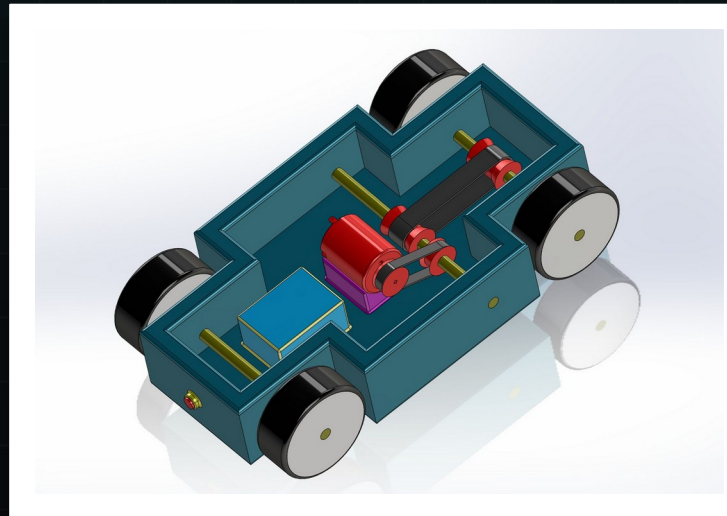
Autonomous Vehicle · Group Project · Oct 2023 – Dec 2023

05

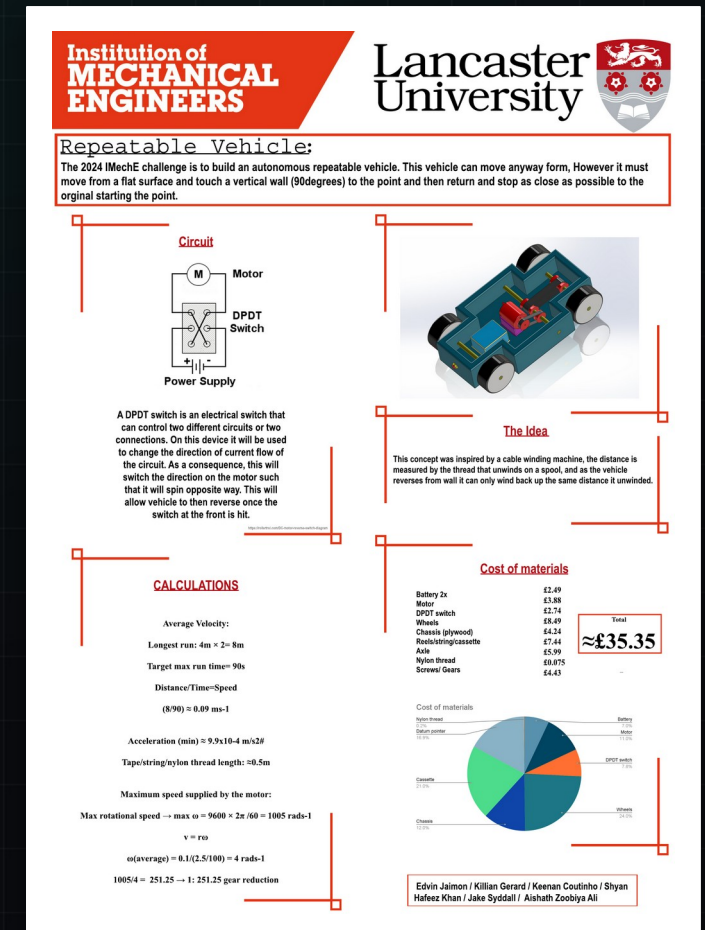
OVERVIEW

A first-year autonomous vehicle challenge solved within a strict £35 budget.

- Designed and built an autonomous vehicle for the 2024 IMechE Foundation Design Challenge, capable of travelling to a vertical wall and returning to its starting point via a DPDT switch-based motor direction reversal circuit.
- Calculated gear reduction ratios and motor specifications to meet target run time and repeatability requirements within a constrained £35 budget.



Vehicle CAD model



Design, circuit & calculations

Hydroponic Plant Growing System

Sustainability Design Project · 2019 – 2020

06

OVERVIEW

An early design project exploring sustainable indoor urban gardening.

- Conceptualised and developed an indoor hydroponic plant growing system to promote sustainable urban gardening, researching urban agriculture techniques and analysing user needs and environmental constraints to inform the design.
- Produced multiple design iterations using sketches and CAD, then prototyped across various materials, iterating with peer feedback to ensure functionality, durability and aesthetic quality.



Prototype growing system